

Revised Load Interconnection System Impact Study Report

For Queue Project Q19-1264

Jamestown 138 kV
August 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1264. The Interconnection Customer proposes a 228.6 MW large load (data center) facility interconnecting at the Jamestown 138 kV substation in the Pennsylvania area. The request was received on August 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 322 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,651,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Jamestown - Barnard 138.0 kV	Beejay - Jenkins Substation 138.0 kV Line	165.3	170	100.0
Boeing Substation - Barnard 138.0 kV	Beejay - Jenkins 138.0 kV	140.6	211	100.0
Ighams - Boeing 138.0 kV line	Kooskia - Warm Springs 138.0 kV	136.3	240	44.9
Warm Springs - Kooskia 138.0 kV line	Inghams - Boeing 138.0 kV	127.7	284	42.2
Jamestown - Kaukauna City Hydro 138.0 kV	Barnard - Boeing 138.0 kV	117.3	226	100.0
Bejay - Kaukauna City Hydro 138.0 kV	Barnard - Boeing 138.0 kV	115.9	245	100.0
Jenkins - Beejy 138.0 kV	Barnard - Boeing 138.0 kV	114.1	259	100.0
Humacao Tc - Dwp 138.0 kV	Boeing Substation - Barnard 138.0 kV	108.4	195	5.2
Humacao Tc Substation - Strasburg 138.0 kV Line	Boeing - Barnard 138.0 kV	104.8	205	5.2
Strasburg - Perryton 138.0 kV	Boeing - Barnard 138.0 kV	104.7	208	5.2
Perryton - Lamar 138.0 kV Line	Barnard - Boeing 138.0 kV	101.4	222	5.2
Lawton E Side - Lamar 138.0 kV	Boeing - Barnard Substation 138.0 kV	100.4	233	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jamestown - Barnard 138.0 kV	Rebuild line with higher-capacity conductor	\$1,558,000	100% (\$1,558,000)
Boeing - Barnrd 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,229,000	100% (\$3,229,000)
Ighams - Boeing 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,776,000	81% (\$11,208,000)
Kooskia - Warm Springs 138.0 kV	Rebuild line with higher-capacity conductor	\$2,781,000	93% (\$2,591,000)
Jamestown - Kaukauna City Hydro 138.0 kV	Rebuild line with higher-capacity conductor	\$6,110,000	100% (\$6,110,000)
Bejay - Kaukauna City Hydro 138.0 kV	Rebuild line with higher-capacity conductor	\$3,874,000	100% (\$3,874,000)
Jenkins - Beejy 138.0 kV	Rebuild line with higher-capacity conductor	\$4,787,000	100% (\$4,787,000)
Dwp - Humacao Tc 138.0 kV	Reconductor line	\$8,058,000	100% (\$8,058,000)
Humacao Tc - Strasburg Substation 138.0 kV	Reconductor line	\$4,902,000	100% (\$4,902,000)
Strasburg Substation - Perryton 138.0 kV L	Reconductor line	\$2,151,000	100% (\$2,151,000)
Perryton - Lamar 138.0 kV	Reconductor line	\$2,210,000	100% (\$2,210,000)
Lawton E Side - Lamar 138.0 kV	Reconductor line	\$3,973,000	100% (\$3,973,000)
	Total allocated to project		\$54,651,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.